GENERAL NOTES

1. EXISTING CONCRETE CONSTRUCTION SHALL BE SCANNED BY X-RAY OR GROUND-PENETRATING RADAR TO IDENTIFY REBAR PRIOR TO CORING OR SAWING. REFER TO ARCHITECTURAL AND STRUCTURAL FOR ADDITIONAL REQUIREMENTS.
2. WORK INCLUDED IN CONTRACT IS SHOWN WITH HEAVIER LINE WEIGHT; WORK SHOWN WITH LIGHT LINE WEIGHT IS (E) TO REMAIN. MAKE MODIFICATIONS TO (E) BRANCH CIRCUITS TO RETAIN CONTINUITY, INCLUDING EQUIPMENT OUTSIDE THE AREA OF WORK.
3. ALL CONDUIT IS TO BE INSTALLED CONCEALED IN FINISHED AREAS UNLESS OTHERWISE NOTED AND SHALL BE A MINIMUM SIZE OF 3/4".
4. COORDINATE REQUIREMENTS AND LOCATIONS OF MECHANICAL EQUIPMENT WITH MECHANICAL DRAWINGS AND MECHANICAL CONTRACTOR PRIOR TO ROUGH-IN.
5. MAKE ALL FINAL ELECTRICAL CONNECTIONS TO EQUIPMENT.
6. FIRE SEAL ALL FIRE RATED WALL AND FLOOR PENETRATIONS.
7. ENSURE THAT NEW CONSTRUCTION DOES NOT BLOCK ACCESS TO EXISTING EQUIPMENT SERVICE CLEARANCES, COMMUNICATION OUTLETS, CABLE TRAYS, PULL BOXES, ETC.
8. CONDUIT ASSOCIATED WITH FEEDERS AND BRANCH CIRCUITS TO BE REMOVED SHALL BE REMOVED IN ITS ENTIRETY BACK TO THE SERVING PANELBOARD.
9. EACH MULTI-WIRE BRANCH CIRCUIT SHALL BE PROVIDED WITH A SEPARATE NEUTRAL FOR EACH BRANCH CIRCUIT.
10. PROTECT ALL EXISTING AUDIO/VISUAL, TELECOM AND DATA OUTLETS AND CABLING DROPS WITHIN THE PROJECT SPACES DURING CONSTRUCTION.
11. THE CONTRACTOR SHALL ENSURE THAT NEW WORK DOES NOT BLOCK ACCESS TO COMMUNICATIONS OUTLETS, CABLE TRAYS, PULLBOXES, AND SIMILAR ITEMS.
12. THE REUSE OF EXISTING CONDUIT/RACEWAY SHALL ONLY BE PERMITTED BY CONSENT OF THE OWNER. THE EXISTING CONDUIT/RACEWAY TO BE REUSED SHALL BE INSPECTED BY THE OWNER'S REPRESENTATIVE PRIOR TO ADDITIONAL WORK BEING DONE TO THAT CONDUIT/RACEWAY.

GENERAL DEMOLITION NOTES

1. WORK SHOWN HATCHED IS TO BE REMOVED; WORK SHOWN WITH LIGHT LINE WEIGHT IS (E) TO REMAIN. MAKE MODIFICATIONS TO (E) BRANCH CIRCUITS TO RETAIN CONTINUITY, INCLUDING EQUIPMENT AND DEVICES OUTSIDE THE AREA OF WORK AND RELOCATED EQUIPMENT AND DEVICES.
2. DEMOLITION DRAWINGS ARE INCLUDED TO GIVE A COMMON BASIS FOR BIDDING. CONTRACTOR IS TO VERIFY (E) CONDITIONS AND REQUIRED DEMOLITION WORK PRIOR TO BID.
3. ALL WIRING, CONDUIT, BOXES AND SUPPORTS NO LONGER REQUIRED SHALL BE COMPLETELY REMOVED FROM THE AREA OF WORK.
4. THE OWNER SHALL HAVE FIRST RIGHT TO REMOVED DEVICES AND EQUIPMENT. IF THE OWNER DOES NOT WANT THE REMOVED DEVICES OR EQUIPMENT, THEN THE CONTRACTOR SHALL DISPOSE OF PROPERLY.
5. FIRE SEAL ALL FIRE RATED WALL AND FLOOR PENETRATIONS.
6. THROUGHOUT THE BUILDING, NO MATTER THE EXTENT OF WORK IN THE PARTICULAR AREA, (E) CABLING, CONDUIT, LUMINAIRES, EQUIPMENT, ETC. SHALL BE PROPERLY SUPPORTED, JUNCTION BOXES COVERED, AND KNOCK OUT OPENINGS COVERED.
7. PRIOR TO DEMOLITION COMPLETELY CIRCUIT TRACE ALL ELECTRICAL SYSTEMS IN THE AREA OF RENOVATION AND SUBMIT INFORMATION, INCLUDING MARKED UP PANELBOARD DIRECTORIES, TO THE ARCHITECT. ADJUST REUSED CIRCUITS AS REQUIRED.
8. WHERE DEVICES AND EQUIPMENT ARE TO BE REMOVED, WALLS SHALL BE PATCHED TO MATCH ORIGINAL FINISH. BLANK COVER PLATES ARE NOT ACCEPTABLE UNLESS OTHERWISE NOTED. PROVIDE BLANK COVER PLATES FOR OUTLET BOXES REMAINING ON (E) WALLS THAT ARE NO LONGER USED.
9. EQUIPMENT MOUNTED ON CEILINGS BEING REMOVED FOR ANY REASON SHALL BE TEMPORARILY SUPPORTED AND INSTALLED ON THE REPLACEMENT CEILING, UNLESS OTHERWISE NOTED.
10. DEMOLITION OF ALL EXISTING TELE/DATA OUTLETS AND LOW VOLTAGE CABLING SHALL BE BY OWNER'S IT VENDOR.

ELECTRICAL LEGEND

(Not all symbols listed below are used on these drawings)

FIRE ALARM

SYMBOL	DESCRIPTION
FACP	FIRE ALARM CONTROL PANEL
FAA	FIRE ALARM ANNUNCIATOR PANEL
EVAC	VOICE EVACUATION CONTROL UNIT
MIC	REMOTE VOICE EVACUATION MICROPHONE
GAP	GRAPHIC ANNUNCIATOR PANEL
ECCU	EMERGENCY COMMUNICATIONS CONTROL UNIT
FSCP	FIRE SUPPRESSION CONTROL PANEL
ARCM	AREA OF REFUGE EMERGENCY COMMUNICATION SYSTEM/MASTER UNIT
ARCR	AREA OF REFUGE EMERGENCY COMMUNICATION SYSTEM/REMOTE UNIT
ESR	ELEVATOR STATUS/RECALL
DACT	DIGITAL ALARM COMMUNICATOR TRANSMITTER
HVAC	CONTROL PANEL FOR VENTILATION, PRESSURIZATION
KB	KNOX BOX
	MANUAL PULL STATION
AIM	ADDRESSABLE INPUT MODULE
AOM	ADDRESSABLE OUTPUT MODULE
WF	SPRINKLER FLOW SWITCH
VS	VALVE SUPERVISORY SWITCH
PS	PRESSURE SWITCH
DH	DOOR HOLDER, MAGNETIC
DCL	DOOR CLOSER
RTS	DUCT SMOKE DETECTOR REMOTE INDICATOR/TEST STATION
	AIR SAMPLING DETECTOR PIPING AND PORT
A	FIRE SERVICE OR EMERGENCY PHONE (* INDICATES DEVICE)
A	ACCESSIBLE
J	JACK
H	HANDSET

SYMBOL	DESCRIPTION
S*	SMOKE DETECTOR (* INDICATES DEVICE)
P	PHOTOELECTRIC
I	IONIZATION
BT	BEAM TRANSMITTER
BR	BEAM RECEIVER
UF	BELOW RAISED ACCESS FLOOR
AS	AIR SAMPLING
SS	SINGLE STATION
S*	DUCT SMOKE DETECTOR (* INDICATES DEVICE)
S	SUPPLY
R	RETURN
P	PHOTOELECTRIC
I	IONIZATION
H*	HEAT DETECTOR/SENSOR (* INDICATES DEVICE)
F	FIXED TEMPERATURE
R	RATE OF RISE
F*	FLAME DETECTOR (* INDICATES DEVICE)
UV	ULTRAVIOLET
IF	INFRARED
	CARBON MONOXIDE DETECTOR
S/H	SMOKE/HEAT DETECTOR/SENSOR COMBINATION
H	AUDIBLE NOTIFICATION HORN
S	AUDIBLE NOTIFICATION SPEAKER
	COMBINATION AUDIBLE/VISIBLE NOTIFICATION HORN/STROBE
	COMBINATION NOTIFICATION SPEAKER/STROBE
cd	VISIBLE FIRE ALARM STROBE (CEILING & WALL MOUNTED)
cd	cd - CANDELA RATING (IF SHOWN)
cd	VISIBLE MASS NOTIFICATION STROBE (CEILING & WALL MOUNTED)
cd	cd - CANDELA RATING (IF SHOWN)
R1	REMOTE INDICATOR LIGHT (CEILING & WALL MOUNTED)
R1	cd - CANDELA RATING (IF SHOWN)
	FIRE ALARM BELL
A	ABORT SWITCH
M	MANUAL RELEASING STATION

COMMUNICATION

SYMBOL	DESCRIPTION
	TELEPHONE TERMINAL BOARD
	VOICE DEVICE
W	WALL
P	PAY
	VOICE DEVICE FLOOR MOUNTED
	TELEVISION DEVICE
B C	BELL/CLOCK DEVICE
WAP	WIRELESS ACCESS POINT
	DATA DEVICE
	DATA DEVICE, FLOOR MOUNTED
	COMBINATION TELEPHONE/DATA DEVICE
	COMBINATION TELEPHONE/DATA DEVICE, FLOOR MOUNTED
	COMMUNICATION/POWER FLOOR BOX, DEVICES AS INDICATED
	COMMUNICATION/POWER POKE THROUGH, DEVICES AS INDICATED
	COMMUNICATION/POWER POLE, DEVICES AS INDICATED

SECURITY AND PUBLIC ADDRESS

SYMBOL	DESCRIPTION
SCP	SECURITY CONTROL PANEL
PAMC	PUBLIC ADDRESS MASTER CONTROL
PAA	PUBLIC ADDRESS AMPLIFIER
P	PUBLIC ADDRESS (* INDICATES DEVICE)
F	FLOOR MOUNTED
B	BELL
BZ	BUZZER
I	INTERCOM STATION
MC	MICROPHONE
PB	PUSHBUTTON
S	SPEAKER
V	VOLUME CONTROL
*	SECURITY DEVICE (* INDICATES DEVICE)
AC	ACCESS CONTROL
C	CAMERA
CR	CARD READER
DR	DOOR RELEASE PUSHBUTTON
DS	DOOR SENSOR
ED	ELECTRIC DOOR STRIKE
DL	ELECTRIC DOOR LATCH
GB	GLASS BREAK SENSOR
H	HORN
KP	KEY PAD
M	MONITOR
MDI	MOTION DETECTOR INFRARED
MDU	MOTION DETECTOR ULTRASONIC
PB	PANIC PUSHBUTTON

NURSE CALL

SYMBOL	DESCRIPTION
NCCP	NURSE CALL CONTROL PANEL
NCPP	NURSE CALL CONSOLE
N*	NURSE CALL: CEILING, WALL MOUNTED (* INDICATES DEVICE)
CB	CORE ZERO (CODE BLUE)
CC	CALL CANCEL
DL	DOME LIGHT
DLZ	DOME LIGHT ZONE
DS	DUTY STATION
EB	EMERGENCY CALL STATION WITH PUSHBUTTON
EF	EMERGENCY CALL STATION WITH FOOT SWITCH
EL	ELAPSE TIMECLOCK
EP	EMERGENCY CALL STATION WITH PULL CORD
ES	EMERGENCY CALL STATION WITH PULL CORD, SHOWER
LCD	SLAVE MASTER STATION DISPLAY
MS	MASTER STATION
P	PATIENT STATION
PB	PUSH BUTTON
PS	PRESENCE STATION
R	REMOTE TIMECLOCK CONTROL
SL	STAFF LOCATER STATION
SS	STAFF STATION

GROUNDING AND LIGHTNING PROTECTION

	INSPECTION WELL
	GROUND ROD
	GROUND BAR
	AIR TERMINAL
	EXOTHERMIC CONNECTION
LP	LIGHTNING PROTECTION BARE CONDUCTOR

REFERENCE SYMBOLS

X	KEY NOTE REFERENCE (DEMOLITION, WHERE APPLICABLE)
	INDICATES DEMOLITION (DASHED or HATCHED)
X	FEEDER REFERENCE
XXXXX	EQUIPMENT REFERENCE (REFER TO EQUIPMENT SCHEDULE)
XX/XXX	DETAIL REFERENCE
	REVISION DELTA
A-1,3,5	BRANCH CIRCUIT HOME RUN: (ALL CIRCUITS HAVE A DEDICATED NEUTRAL)
	ARROWS INDICATE NUMBER OF CIRCUITS
	TEXT INDICATES PANELBOARD CIRCUIT
PNL #	CIRCUIT INDICATION FOR ALL DEVICES WITHIN AN AREA OR ROOM, OCCASIONALLY A CIRCUIT NUMBER IS ADJACENT FOR CLARITY. (ALL CIRCUITS HAVE A DEDICATED NEUTRAL.)

LIGHTING

SYMBOL	DESCRIPTION
	SURFACE LUMINAIRE
	RECESSED LUMINAIRE
OR	EMERGENCY OPERATION
OR	CRITICAL OPERATION
	STRIP LUMINAIRE
	SURFACE CEILING MOUNTED LUMINAIRE
	RECESSED CEILING MOUNTED LUMINAIRE
	WALL MOUNTED LUMINAIRE
	COVE LIGHT/UNDERCOUNTER LUMINAIRE
	RECESSED WALL LUMINAIRE
	RECESSED WALL WASH LUMINAIRE
	SURFACE WALL WASH LUMINAIRE
	DARKROOM SAFE LIGHT, AS INDICATED
	EMERGENCY BATTERY PACK UNIT
	SINGLE FACE EXIT SIGN WITH DIRECTIONAL ARROWS
	DOUBLE FACE EXIT SIGN WITH DIRECTIONAL ARROWS
	POLE MOUNTED LUMINAIRE
	FLOOD LIGHT
	TRACK LIGHTING
A	UPPER CASE DESIGNATES LUMINAIRE TYPE
o	SUBSCRIPT INDICATES SWITCHLEG

POWER

SYMBOL	DESCRIPTION
	SINGLE RECEPTACLE
IG	DUPLEX RECEPTACLE (ESSENTIAL POWER)
	ISOLATED GROUND
	DOUBLE DUPLEX RECEPTACLE (ESSENTIAL POWER)
	DUPLEX RECEPTACLE, CEILING MOUNTED (ESSENTIAL POWER)
	DOUBLE DUPLEX RECEPTACLE, CEILING MOUNTED (ESSENTIAL POWER)
	DUPLEX RECEPTACLE, FLOOR MOUNTED (ESSENTIAL POWER)
	DOUBLE DUPLEX RECEPTACLE, FLOOR MOUNTED (ESSENTIAL POWER)
GFCI	DUPLEX RECEPTACLE, WITH GFCI PROTECTION
WP	WEATHER PROOF (WEATHERPROOF WHILE IN-USE)
	SPECIAL DUPLEX RECEPTACLE
USB	TWO INTEGRAL USB PORTS
SW	HALF SWITCHED
	SPECIAL RECEPTACLE: WALL, FLOOR, CEILING
	COMMUNICATION/POWER FLOOR BOX, DEVICES AS INDICATED
	COMMUNICATION/POWER POKE THROUGH, DEVICES AS INDICATED
	COMMUNICATION/POWER POLE, DEVICES AS INDICATED
B B	BUSWAY
M#	MULTI-OUTLET ASSEMBLY
	# INDICATES DEVICE SPACING ON-CENTER
	EXISTING PANELBOARD
	NEW PANELBOARD OR NEW LOCATION
T	TRANSFORMER
M	UTILITY METER
	NON-FUSED DISCONNECT SWITCH
F	FUSED DISCONNECT SWITCH
CB	CIRCUIT BREAKER
VFD	VARIABLE FREQUENCY DRIVE
	MOTOR STARTER
	COMBINATION STARTER/DISCONNECT
M	MOTOR
T	PAD MOUNTED TRANSFORMER
	DISTRIBUTION EQUIPMENT

SWITCHING

SYMBOL	DESCRIPTION
S (S)	SINGLE POLE SWITCH (LOW VOLTAGE SWITCH)
o	SWITCHING ZONE
2	TWO POLE SWITCH
3	THREE-WAY SWITCH
4	FOUR-WAY SWITCH
K	KEY OPERATED SWITCH
M(TO)	MOTOR RATED SWITCH (THERMAL OVERLOAD)
D	DIMMER SWITCH
EP	EXPLOSION PROOF SWITCH
MC	MOMENTARY CONTACT SWITCH
OS	OCCUPANCY SENSOR
P	SWITCH with ILLUMINATED HAND (LOAD OFF)
PI	SWITCH with PILOT LOGHT (LOAD ON)
VS	VARIABLE SPEED SWITCH
OS (OS)	OCCUPANCY SENSOR (VACANCY SENSOR)
A	UPPER CASE DESIGNATES TYPE
o	SUBSCRIPT INDICATES SWITCH LEG
	PUSHBUTTON
EPO	EMERGENCY POWER OFF
PAD	POWER ASSIST DOOR
OHD	OVERHEAD DOOR
x	LIGHTING CONTACTOR (REFERENCE SCHEDULE)
P*	PHOTOCELL CONTROL
TC	TIMECLOCK

RACEWAYS

SYMBOL	DESCRIPTION
	CONDUIT
UG	CONDUIT, UNDERGROUND(UG) OR UNDERFLOOR(UF)
	J-HOOK SYSTEM
	CABLE TRAY
PB	PULL BOX
	INDICATES DEMOLITION
OR	CONDUIT TURNED DOWN OR UP
	FLEXIBLE CONNECTION
	BUSHED CONDUIT
OR	CONDUIT CAP OR BUSHED CONDUIT WITH CONDUCTOR
	SEAL-OFF
OR	JUNCTION BOX, WALL OR CEILING MOUNTED (FLOOR MOUNTED)

ONE-LINE DIAGRAM SYMBOLS

SYMBOL	DESCRIPTION
	DISCONNECT SWITCH
	DISCONNECT SWITCH, FUSED
	CIRCUIT BREAKER
	FUSE
	GROUND
	CURRENT TRANSFORMER
	POTENTIAL TRANSFORMER
	WEATHERHEAD
1	SHORT CIRCUIT CURRENT NODE
	CONTACT, NORMALLY OPEN
	CONTACT, NORMALLY CLOSED
LB	TERMINATIONS
	LOAD BREAK
NLB	NO LOAD BREAK
#	ANSI PROTECTIVE DEVICE
	DRAW-OUT DEVICE
OR	DISCONNECT SWITCH, F INDICATES FUSED
CB	ENCLOSED CIRCUIT BREAKER
VFD	VARIABLE FREQUENCY DRIVE
T #	TRANSFORMER
T	PAD MOUNTED TRANSFORMER
A	PANELBOARD
DM	DIGITAL METER
VS	VOLTMETER TEST SWITCH
AS	AMMETER TEST SWITCH
V	VOLTMETER
A	AMMETER
	FEEDER REFERENCE
E G	ENGINE GENERATOR
ATS MTS	TRANSFER SWITCH
	AUTOMATIC
	MANUAL
GFP	GROUND FAULT PROTECTION
SPD-x	SURGE PROTECTIVE DEVICE (REFERENCE SCHEDULE)
EGAP	ENGINE GENERATOR ANNUNCIATOR PANEL
M	METER
	COMBINATION STARTER/DISCONNECT
	MOTOR STARTER
M	MOTOR
	EQUIPMENT ENCLOSURE

ELECTRICAL DRAWING INDEX

SHEET NO.	SHEET TITLE
E000	ELECTRICAL LEGENDS AND GENERAL NOTES
E200	PARTIAL ELECTRICAL PLAN - BASEMENT
E201	ELECTRICAL PLANS - GROUND FLOOR
ED401	ELECTRICAL
ED402	ELECTRICAL
ED403	POWER AND SYSTEMS DEMOLITION PLAN - SECOND FLOOR
E401	POWER AND SYSTEMS PLAN - BASEMENT AND GROUND FLOOR
E403	POWER AND SYSTEMS PLAN - FIRST FLOOR
E404	POWER AND SYSTEMS PLAN - SECOND FLOOR
E601	ELECTRICAL DETAILS

ABBREVIATIONS

A	AMPERES	NIC	NOT IN CONTRACT
ac	ABOVE COUNTER	NTS	NOT TO SCALE
AFF	ABOVE FINISHED FLOOR	(N)	NEW
AFG	ABOVE FINISHED GRADE	NC	NORMALLY CLOSED
ATS	AUTOMATIC TRANSFER SWITCH	NO	NORMALLY OPEN
BFG	BELOW FINISHED GRADE	NL	NIGHT LIGHT
C	CONDUIT	OC	ON CENTER
CATV	CABLE TELEVISION	OHD	OVERHEAD DOOR
CCTV	CLOSED CIRCUIT TELEVISION	OHP	OVERHEAD PROJECTOR
CB	CIRCUIT BREAKER	PVC	POLY VINYL CHLORIDE
(D)	DEMOLISH & REMOVE	(R)	RELOCATED
(E)	EXISTING	(RR)	REMOVE & RELOCATE
E/G	ENGINE GENERATOR	RAF	RAISED ACCESS FLOOR
EM	EMERGENCY	RMC	RIGID METAL (STEEL) CONDUIT
EMT	ELECTRICAL METALLIC TUBING	RGS	RIGID GALVANIZED STEEL CONDUIT
EP	EXPLOSION PROOF	SB	STAND-BY
EWC	ELECTRIC WATER COOLER	SPD	SURGE PROTECTIVE DEVICE
(F)	FUTURE	TP	TAMPER PROOF
FA	FIRE ALARM	TYP	TYPICAL
G	GROUND	UF	UNDER FLOOR
GFCI	GROUND FAULT CIRCUIT INTERRUPTER	UG	UNDER GROUND
GFI	GROUND FAULT INDICATION	UNO	UNLESS NOTED OTHERWISE
GFP	GROUND FAULT PROTECTION	UPS	UNINTERRUPTABLE POWER SUPPLY
HOA	HAND "OFF" AUTOMATIC	V	VOLTS
IG	ISOLATED GROUND	VA	VOLTS AMPERES
KVA	KILOVOLT AMPERES	VFD	VARIABLE FREQUENCY DRIVE
KW	KILOWATTS	W/	WITH
MCB	MAIN CIRCUIT BREAKER	W/O	WITHOUT
MCC	MOTOR CONTROL CENTER	WG	WIREGUARD
MLO	MAIN LUGS ONLY	WP	WEATHER PROOF
MV	MEDIUM VOLTAGE	XFMR	TRANSFORMER

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UNIVERSITY OF NORTHERN COLORADO (UNC)
STATE PROJECT # 2017-049P22
GRAY HALL MECHANICAL SYSTEMS REPLACEMENT & LIFE SAFETY UPGRADES
1813 8th Ave., Greeley, Co 80631

SBEC Project #:	230026
Scale:	AS SHOWN
Drawn By:	
Designed By:	AR
Checked By:	JE

Issued For:	Date:
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ELECTRICAL LEGENDS AND GENERAL NOTES

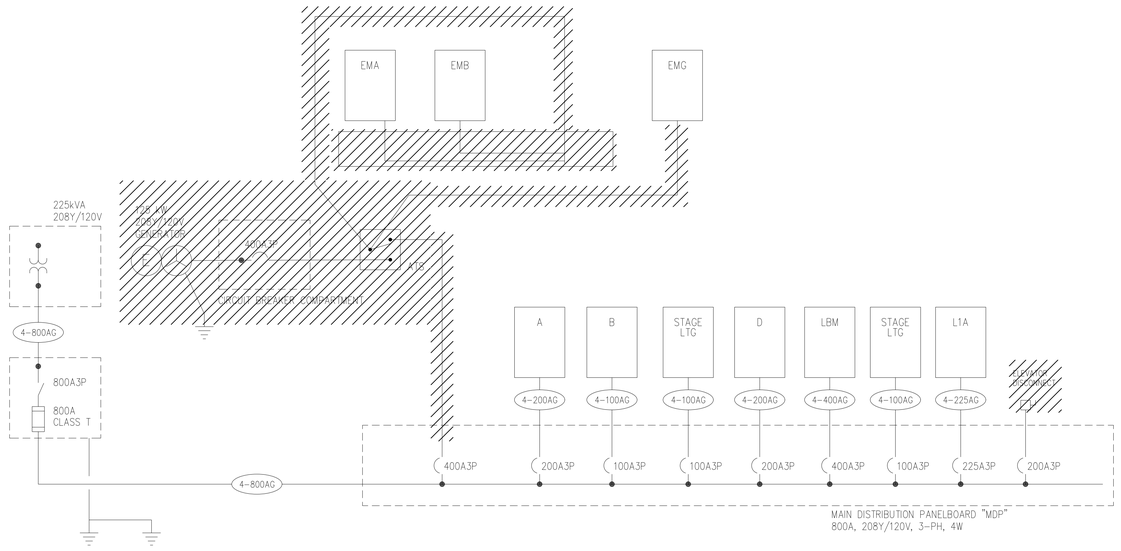
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KEY NOTES

1. NOT USED.



ELECTRICAL ONE-LINE DIAGRAM - DEMOLITION

SCALE: NONE

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ELECTRICAL ONE-LINE DIAGRAM - DEMOLITION

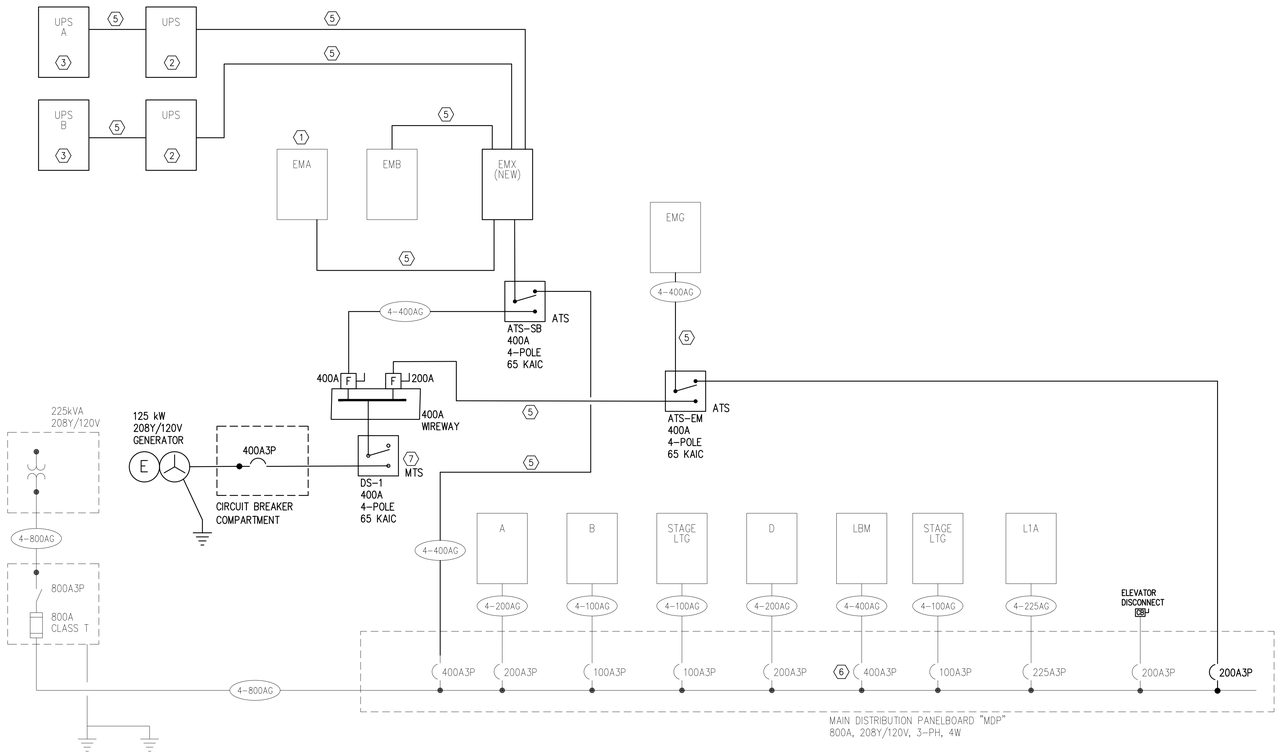
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KEY NOTES

1. EXISTING PANEL "EMA" TO REMAIIN. RE-FEED EXISTING PANEL FROM NEW PANEL "EMX" VIA NEW GUTTER.
2. NEW STAND-ALONE UPS UNIT IN DATA CENTER. EATON 9355 30 KVA WITH MAINTENANCE BYPASS.
3. NEW UPS BRANCH DISTRIBUTION PANEL TO BE MOUNTED NEAR UPS IN DATA CENTER.
4. NEW FEEDER FROM NEW PANEL "EMX"(NEW) TO EXISTING PANELS "EMA" AND "EMB"..
5. NEW FEEDERS AND EMT CONDUIT.
6. NEW LABEL FOR CIRCUIT THAT MATCHES NAME OF PANEL, "PANEL LBM".
7. PROVIDE DUAL PURPOSE DOCKING STATION FOR CONNECTION OF TEMPORARY GENERATOR AS REQUIRED BY CODE AND FOR CONNECTION OF LOAD BANK..



ELECTRICAL ONE-LINE DIAGRAM - NEW WORK
SCALE: NONE

-

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ELECTRICAL ONE-LINE DIAGRAM - NEW WORK

E011

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EQUIPMENT SCHEDULE

DESIGNATION	DESCRIPTION	ELECTRICAL CHARACTERISTICS						LOCAL DISCONNECT		BRANCH CIRCUIT OR FEEDER	PANELBOARD/ DISTRIBUTION BOARD	REMARKS
		HP	KVA	FLA	MCA	VOLTAGE	PHASE	SWITCH	FUSE			
AHU-1(a)	AIR HANDLER	15	17.40	48.3	--	208	3	100A3P	70.0A FRN-R	3-70AG		
AHU-1(b)	AIR HANDLER (LIGHTING)	--	0.31	2.61	--	120	1	--	--	2-20AG		
AHU-1(c)	AIR HANDLER (RECEPTACLE)	--	0.96	8	--	120	1	--	--	2-20AG		
DOAS-1	AIR HANDLER	10	11.60	32.2	--	208	3	60A3P	50.0A FRN-R	3-50AG		
FCU-C1100	FAN COIL UNIT	1/8	0.47	3.9	--	120	1	SMTO	--	2-20AG		
FCU-1104A	FAN COIL UNIT	1/4	0.70	5.8	--	120	1	SMTO	--	2-20AG		
FCU-1104B	FAN COIL UNIT	1/4	0.70	5.8	--	120	1	SMTO	--	2-20AG		
FCU-1102	FAN COIL UNIT	1/8	0.47	3.9	--	120	1	SMTO	--	2-20AG		
FCU-1106	FAN COIL UNIT	1/4	0.70	5.8	--	120	1	SMTO	--	2-20AG		
FCU-1108	FAN COIL UNIT	1/4	0.70	5.8	--	120	1	SMTO	--	2-20AG		
FCU-R1109	FAN COIL UNIT	1/8	0.47	3.9	--	120	1	SMTO	--	2-20AG		
FCU-1203	FAN COIL UNIT	1/4	0.70	5.8	--	120	1	SMTO	--	2-20AG		
FCU-1202	FAN COIL UNIT	1/8	0.47	3.9	--	120	1	SMTO	--	2-20AG		
FCU-1308	FAN COIL UNIT	1-1/2	2.49	6.9	--	208	3	30A3P	10.0A FRN-R	3-20AG		
FCU-R1314	FAN COIL UNIT	1/2	0.90	2.5	--	208	3	30A3P	4.0A FRN-R	3-20AG		
FCU-Q1306	FAN COIL UNIT	1/2	0.90	2.5	--	208	3	30A3P	4.0A FRN-R	3-20AG		
FCU-L1300	FAN COIL UNIT	1/2	0.90	2.5	--	208	3	30A3P	4.0A FRN-R	3-20AG		
FCU-1301C	FAN COIL UNIT	1/4	0.70	5.8	--	120	1	SMTO	--	2-20AG		
FCU-S1400	FAN COIL UNIT	1/4	0.70	5.8	--	120	1	SMTO	--	2-20AG		
FCU-2102	FAN COIL UNIT	1/8	0.47	3.9	--	120	1	SMTO	--	2-20AG		
FCU-2104	FAN COIL UNIT	1/4	0.70	5.8	--	120	1	SMTO	--	2-20AG		
FCU-2101	FAN COIL UNIT	1/3	0.86	7.2	--	120	1	SMTO	--	2-20AG		
FCU-2100C	FAN COIL UNIT	1/4	0.70	5.8	--	120	1	SMTO	--	2-20AG		
FCU-2100B	FAN COIL UNIT	1/4	0.70	5.8	--	120	1	SMTO	--	2-20AG		
FCU-2100A	FAN COIL UNIT	1/8	0.47	3.9	--	120	1	SMTO	--	2-20AG		
FCU-L2100	FAN COIL UNIT	1/8	0.47	3.9	--	120	1	SMTO	--	2-20AG		
FCU-2206D	FAN COIL UNIT	1/4	0.70	5.8	--	120	1	SMTO	--	2-20AG		
FCU-2206	FAN COIL UNIT	1/4	0.70	5.8	--	120	1	SMTO	--	2-20AG		
FCU-2206A	FAN COIL UNIT	1/8	0.47	3.9	--	120	1	SMTO	--	2-20AG		
FCU-2206B	FAN COIL UNIT	1/8	0.47	3.9	--	120	1	SMTO	--	2-20AG		
FCU-2200A	FAN COIL UNIT	1/4	0.70	5.8	--	120	1	SMTO	--	2-20AG		
FCU-2200B	FAN COIL UNIT	1/8	0.47	3.9	--	120	1	SMTO	--	2-20AG		
FCU-2301A	FAN COIL UNIT											
FCU-2301B	FAN COIL UNIT	1	1.73	4.8	--	208	3	30A3P	7.0A FRN-R	3-20AG		
FCU-2301E	FAN COIL UNIT	1/4	0.70	5.8	--	120	1	SMTO	--	2-20AG		
FCU-2301A	FAN COIL UNIT	1	1.73	4.8	--	208	3	30A3P	7.0A FRN-R	3-20AG		
FCU-2305B	FAN COIL UNIT	1/8	0.47	3.9	--	120	1	SMTO	--	2-20AG		
FCU-C2300	FAN COIL UNIT											
FCU-2305A	FAN COIL UNIT	1/2	0.90	2.5	--	208	3	30A3P	4.0A FRN-R	3-20AG		
FCU-3102	FAN COIL UNIT	1/2	0.90	2.5	--	208	3	30A3P	4.0A FRN-R	3-20AG		
FCU-3101	FAN COIL UNIT	1/2	0.90	2.5	--	208	3	30A3P	4.0A FRN-R	3-20AG		
EF-1	EXHAUST FAN	1/2	0.90	2.5	--	208	3	30A3P	4.0A FRN-R	3-20AG		
EF-2	EXHAUST FAN	1/2	0.90	2.5	--	208	3	30A3P	4.0A FRN-R	3-20AG		
EF-3	EXHAUST FAN											
GF-1	GLYCOL FEEDER											
CHWP-1	CHILLED WATER PUMP	5	6.30	17.5	--	208	3	30A3P	30.0A FRN-R	3-30AG		
CHWP-2	CHILLED WATER PUMP	5	6.30	17.5	--	208	3	30A3P	30.0A FRN-R	3-30AG		
HWP-1	HEATING WATER PUMP	3	3.96	11.0	--	208	3	30A3P	17.5A FRN-R	3-20AG		
HWP-2	HEATING WATER PUMP	3	3.96	11.0	--	208	3	30A3P	17.5A FRN-R	3-20AG		
AHU-1-HCP-1	HEATING COIL CIRC PUMP	1/12	0.36	3.0	--	120	1	30A1P	4.5A FRN-R	2-20AG		
DOAS-1-HCP-1	HEATING COIL CIRC PUMP	1/12	0.36	3.0	--	120	1	30A1P	4.5A FRN-R	2-20AG		
RTU-2-HCP-1	HEATING COIL CIRC PUMP	1/75	0.10	0.9	--	120	1	30A1P	1.4A FRN-R	2-20AG		
RTU-2	ROOFTOP UNIT	5	6.30	17.5	--	208	3	30A3P	30.0A FRN-R	3-30AG		
TCP-X	TEMPERATURE CONTROL PANEL		86.53									

GENERAL REMARKS:

A VERIFY THE INFORMATION SCHEDULED WITH MECHANICAL CONTRACTOR AND SUBMITTALS, INFORM DESIGN TEAM OF ANY DEVIATION.

B COORDINATE LOCATION OF POINTS OF CONNECTION WITH EQUIPMENT SUPPLIED PRIOR TO ROUGH-IN.

C PROVIDE APPROPRIATE NEMA RATED ENCLOSURE BASED UPON THE LOCATION OF ELECTRICAL COMPONENTS.

D

E

SPECIFIC REMARKS:

1 MECHANICAL CONTRACTOR SUPPLIED VFD INSTALLED BY ELECTRICAL CONTRACTOR. PROVIDE LOCAL DISCONNECT WITH NORMALLY OPEN EARLY BREAK AUXILIARY CONTACTS FOR OVERRIDE OF VFD. CONNECT AUXILIARY CONTACTS TO THE VFD VIA 2#12, 3/4"C TO INDICATE THAT THE LOCAL DISCONNECT IS OPEN AND TO DISABLE OPERATION OF THE VFD.

2 MECHANICAL CONTRACTOR SUPPLIED STARTER INSTALLED BY ELECTRICAL CONTRACTOR.

3 CONNECT VFD LINE FILTER UNIT TO VFD VIA 2#12, 3/4"C TO DISABLE OPERATION OF THE LINE FILTER CAPACITORS WHEN VFD IS OPERATING AT LOW-LOAD OR NO LOAD.

4 PROVIDE CONTROL SWITCH ABOVE COUNTER.

5 CONNECTED WITH ROOM LIGHTING [REFER TO DETAIL].

6 SEPARATE 120 VOLT CONNECTION TO UNIT LIGHTING.

7 FAN WALL UNIT [WITH MULTIPLE DISCONNECTS].

8 UNIT CONNECTED TO STAND-BY POWER.

9 UNIT CONNECTED TO EMERGENCY POWER.

10 FULLY REDUNDANT DUPLEX UNIT; ONLY ONE MOTOR OPERATES AT A TIME.

11 LEAD, STAND-BY, BOTH PUMPS DO NOT OPERATE CONCURRENTLY.

12 SPECIFIED DUPLEX EQUIPMENT IS A SINGLE POINT CONNECTION, IF EQUIPMENT PROVIDED REQUIRES TWO POINTS OF CONNECTION, MECHANICAL CONTRACTOR IS RESPONSIBLE FOR THE CORRESPONDING ELECTRICAL CHANGES.

13 PROVIDE TWO LOCAL DISCONNECT SWITCHES; ONE FOR TOTAL UNIT AND ONE FOR FAN WALL. ITEMS LISTED ON THE TOP LINE ARE FOR THE TOTAL UNIT.

14 CONNECT MOTORIZED DAMPER.

15 LOCAL MANUAL MOTOR STARTER FURNISHED WITH UNIT.

16 HEAT TRACE BRANCH CIRCUIT TO BE FED FROM 30MA GROUND FAULT EQUIPMENT PROTECTION (GFEP) TYPE CIRCUIT BREAKER AT PANELBOARD INDICATED (OR PROVIDE GFEP AT HEAT TRACE CONTROL PANEL).

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UNIVERSITY OF NORTHERN COLORADO (UNC)
STATE PROJECT # 2017-049P22
GRAY HALL MECHANICAL SYSTEMS REPLACEMENT & LIFE SAFETY UPGRADES
1813 8th Ave., Greeley, Co 80631

SBEC Project #: 230026
Scale: AS SHOWN
Drawn By:
Designed By: AR
Checked By: JE

Issued For: Date:
100% DD 03/11/2025

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NOTE:

EXISTING CONDITIONS ARE SHOWN WITH LIGHT LINE WEIGHT.

NEW WORK INCLUDED IN THIS CONTRACT IS SHOWN WITH HEAVY LINE WEIGHT.

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ELECTRICAL SCHEDULES

E020

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PANELBOARD: "LBM"

MANUFACTURER AND TYPE: SIEMENS ITE CDP-7 | X NEUTRAL | TVSS
NEW | X GROUND | ISO-GROUND
X EXISTING | MAIN CB | RATING: 400 AMPERES
BUS RATING = 400 AMPERES | X MLO
AIC RATING = AMPERES
VOLTAGE = 208 Y/120V, 3 PHASE, 4 WIRE | X SURFACE | RECESSED

TYP	VA	LOAD DESCRIPTION	P	CB	CCT	PH	CCT	CB	P	LOAD DESCRIPTION	VA	TYP
		RETURN PUMP	3	20	1	A	2	30	3	FAN		
		OPEN			3	B	4			--		
		SUMP PUMP			5	C	6			--		
		ON LOBBY PAY	1	20	7	A	8	30	3	AIR COMP.		
		CANDY MACH	1	20	9	B	10			--		
		MENS+WOMENS W RR EXHAUST FAN	1	20	11	C	12			--		
			1	30	13	A	14	50	2			
		TRIDIUM CABNET	1	20	15	B	16			--		
		ICE MAKER	1	30	17	C	18	30	2	208V 30A OUTLET ON STAIRWAY "OFF"		
		BACK FLOW PREVENTER HEAT TRACE	1	20	19	A	20			--		
		SPACE			21	B	22	20	3	HEAT PUMP		
		SPACE			23	C	24			--		
		SPACE			25	A	26			--		
		SPACE			27	B	28			SPACE		
		SPACE			29	C	30			SPACE		
					31	A	32					
					33	B	34					
					35	C	36					
					37	A	38					
					39	B	40					
					41	C	42					

CONNECTED LOAD: A = VA; A TO B = %; B = VA; B TO C = %; C = VA; C TO A = %

LOAD TYPE: E = EQUIPMENT; H = HEATING; K = KITCHEN EQUIPMENT; L = LIGHTING; M = MOTOR; P = PANELBOARD; R = RECEPTACLE; T = TRANSFORMER

PHASE LOADING AND BALANCE

		CONNECTED (VA)	DEMAND FACTOR	DEMAND LOAD (VA)	NEC DEMAND	NEC DEMAND LOAD (VA)
LIGHTING (L)		-	1.0	-	1.25	-
REC	1st 10 KVA	-	1.0	-	1.0	-
(R)	REMAINING	-	1.0	-	0.5	-
LARGEST MOTORS (M)		-	1.0	-	1.25	-
REMAINING MOTORS (M)		-	1.0	-	1.0	-
EQUIPMENT (E)		-	1.0	-	1.0	-
HEATING (H)		-	1.0	-	1.25	-
KITCHEN (K)		-	1.0	-	1.00	-
PANEL and/or XFMR (P)(T)		-	1.0	-	1.0	-
TOTAL		-		-		-

NEC DEMAND LOAD = A
SPARE CAPACITY = 400 A 100.0 %
TOTAL AVAILABLE = 400 A

NOTES:

PANELBOARD: "EMG" (NOTE 2)

MANUFACTURER AND TYPE: ITE – CDP–7 | X NEUTRAL | TVSS
NEW | X GROUND | ISO–GROUND
X EXISTING | X MAIN CB | RATING: 100 AMPERES
BUS RATING = 100 AMPERES | MLO
AIC RATING = 10,000 AMPERES
VOLTAGE = 208 Y/120V, 3 PHASE, 4 WIRE | X SURFACE | RECESSED

TYP	VA	LOAD DESCRIPTION	P	CB	CCT	PH	CCT	CB	P	LOAD DESCRIPTION	VA	TYP	
		SPARE	1	20	1	A	2	20	1	SPARE			
		SPARE	1	20	3	B	4	20	1	RELEASE PANEL Q1206		E	
R		SOUTHEAST OFFICE	1	20	5	C	6	20	1	SPARE			
		SPARE	1	20	7	A	8	20	1	SPARE			
R		NORTHEAST OFFICE	1	20	9	B	10	20	1	OFFICE IN DISPATCH		R	
4 E	1000	FACP	1	20	11	C	12	20	1	DRY–PIPE COMPRESSOR	1176	M	1,4
		SPARE	1	20	13	A	14	20	1	DRY–PIPE COMPRESSOR	1176	M	1,4
		SPARE	1	20	15	B	16	20	1	DRY–PIPE COMPRESSOR	1176	M	1,4
		SPARE	1	20	17	C	18	20	1	SPARE			
		SPARE	1	20	19	A	20	20	1	SPARE			
		SPARE	1	20	21	B	22	20	1	SPARE			
		SPARE	1	20	23	C	24	20	1	TRIDIUM PANEL		E	
		SPARE	1	20	25	A	26	20	1	REC–DISPATCH DESK		R	
		SPARE	1	20	27	B	28	20	1	REC–DISPATCH DESK		R	
		SPARE	1	20	29	C	30	20	1	REC–DISPATCH DESK		R	
E		6–K EAST	2	30	31	A	32	20	1	SPARE			
E		6–K EAST			33	B	34	20	1	SPARE			
R		REC–E WALL S RECTIFIER	1	15	35	C	36	20	2	OUTSIDE COOLING FAN		E	
E		DISPATCH COPIER	1	20	37	A	38			OUTSIDE COOLING FAN		E	
E		HONEYWELL BOX N.E. CORNER	1	20	39	B	40	20	2	BACKUP UPS		E	
R		REC–E WALL S.	1	20	41	C	42			BACKUP UPS		E	

CONNECTED LOAD: A = 1,176 VA; A TO B = 100 %; B = 1,176 VA; B TO C = 54 %; C = 2,176 VA; C TO A = 54 %

LOAD TYPE: E = EQUIPMENT; H = HEATING; K = KITCHEN EQUIPMENT; L = LIGHTING; M = MOTOR; P = PANELBOARD; R = RECEPTACLE; T = TRANSFORMER

PHASE LOADING AND BALANCE

		CONNECTED (VA)	DEMAND FACTOR	DEMAND LOAD (VA)	NEC DEMAND	NEC DEMAND LOAD (VA)
LIGHTING (L)		–	1.0	–	1.25	–
REC	1st 10 KVA	–	1.0	–	1.0	–
(R)	REMAINING	–	1.0	–	0.5	–
LARGEST MOTORS (M)		1,176	1.0	1,176	1.25	1,470
REMAINING MOTORS (M)		2,352	1.0	2,352	1.0	2,352
EQUIPMENT (E)		1,000	1.0	1,000	1.0	1,000
HEATING (H)		–	1.0	–	1.25	–
KITCHEN (K)		–	1.0	–	1.00	–
PANEL and/or XFMR (P)(T)		–	1.0	–	1.0	–
TOTAL		4,528		4,528		4,822

NEC DEMAND LOAD = 13 A
SPARE CAPACITY = 87 A 86.6 %
TOTAL AVAILABLE = 100 A

NOTES:
1 NEW LOAD ON EXISTING CIRCUIT BREAKER.
2. PANEL WILL BE METERED TO COMPLETE LOAD STUDY.
3 REVISED LOAD ON EXISTING CIRCUIT BREAKER.
4 LOCK ON CIRCUIT BREAKER.

PANELBOARD: "A"

MANUFACTURER AND TYPE: SIEMENS ITE CDP-7 | X NEUTRAL | TVSS
NEW | X GROUND | ISO-GROUND
X EXISTING | MAIN CB | RATING: 225 AMPERES
BUS RATING = 225 AMPERES | X MLO
AIC RATING = AMPERES
VOLTAGE = 208 Y/120V, 3 PHASE, 4 WIRE | X SURFACE | RECESSED

TYP	VA	LOAD DESCRIPTION	P	CB	CCT	PH	CCT	CB	P	LOAD DESCRIPTION	VA	TYP
		WHEEL CHAIR LIFT	1	20	1	A	2	100	3	SPARE		
		RESTROOM LTNG. & EXHAUST	1	20	3	B	4			PATROL SPRVSR & SECTRY REC.		
			1	20	5	C	6			CONF. RM. LTNG.		
		WEST HALL LTNG.	1	20	7	A	8	20	1	CONF. RM. LTG		
		STOREROOM LTNG.	1	20	9	B	10	20	1	INTEROG RMS.		
			1	20	11	C	12	20	1	LTNG. E. STOR. RM., REC. E. & W. SOTRAGE & COMP. MAINT		
		COMPUTER RM. LTNG.	1	20	13	A	14	20	1	SERVICE COUNTER REC 1ST FLOOR		
		N.E. OFFICE LTNG.	1	20	15	B	16	20	1	E. OFFICE REC.		
		LOBBY, INFO, EXT. FRONT LTNG.	1	20	17	C	18	20	1	N.E. OFFICE & LOBBY REC.		
		N. & E. BULL PEN LTNG.	1	20	19	A	20	20	1	BOOTH OFFICES REC. 1ST FLOOR		
			1	20	21	B	22	20	1	SERVICE COUNTER W. REC.		
		INNER OFFICES IN WRK. CNTRL & COMM. CTR. LTNG.	1	20	23	C	24	20	1	INFO BOOTH REC.		
			1	20	25	A	26	20	1	COMP.* TYPE BENCH WRK. CNTRL REC.		
		OFFICE MNGR. LTNG., REC. SO BULL PEN & PARKING DECK	1	20	27	B	28	20	1	COMP.*		
			1	20	29	C	30	20	1	COMP.*		
		SO. BULLPEN LTNG.	1	20	31	A	32	20	1	COPY MACHINE		
		SPARE TO PH. RM.	1	20	33	B	34	20	1	COMP.*		
		RM 2102 LTG + RCPT	1	20	35	C	36	20	1	COMP.* MICRO DATA		
			1	20	37	A	38	20	1	WOMEN'S R.R. REC.		
		CUSTODIAL OFFICE	1	20	39	B	40	20	2	2101 AC		
		BLANK	1	20	41	C	42					

CONNECTED LOAD: A = VA; A TO B = %; B = VA; B TO C = %; C = VA; C TO A = %

LOAD TYPE: E = EQUIPMENT; H = HEATING; K = KITCHEN EQUIPMENT; L = LIGHTING; M = MOTOR; P = PANELBOARD; R = RECEPTACLE; T = TRANSFORMER

PHASE LOADING AND BALANCE

		CONNECTED (VA)	DEMAND FACTOR	DEMAND LOAD (VA)	NEC DEMAND	NEC DEMAND LOAD (VA)
LIGHTING (L)		-	1.0	-	1.25	-
REC	1st 10 KVA	-	1.0	-	1.0	-
(R)	REMAINING	-	1.0	-	0.5	-
LARGEST MOTORS (M)		-	1.0	-	1.25	-
REMAINING MOTORS (M)		-	1.0	-	1.0	-
EQUIPMENT (E)		-	1.0	-	1.0	-
HEATING (H)		-	1.0	-	1.25	-
KITCHEN (K)		-	1.0	-	1.00	-
PANEL and/or XFMR (P)(T)		-	1.0	-	1.0	-
TOTAL		-		-		-

NEC DEMAND LOAD = A
SPARE CAPACITY = 225 A 100.0 %
TOTAL AVAILABLE = 225 A

NOTES:

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STATE PROJECT # 2017-049P22
GRAY HALL MECHANICAL SYSTEMS REPLACEMENT & LIFE SAFETY UPGRADES
1813 8th Ave., Greeley, Co 80631

SBEC Project #: 230026
Scale: AS SHOWN
Drawn By:
Designed By: AR
Checked By: JE

Issued For: Date:
100% DD 03/11/2025

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ELECTRICAL PANELBOARD SCHEDULES - EMERGENCY

E040

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GENERAL NOTES

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DEMOLITION KEY NOTES

1. DEMOLISH EXISTING FACP ONCE THE NEW FACP HAS BEEN INSTALLED. REUSE 120v POWER CIRCUIT SERVING EXISTING FACP FOR THE NEW FACP.
2. (E) GENERATOR TO BE REPLACED.
3. (E) ATS TO BE REMOVED AND REPLACED WITH NEW ATS'S.



FIRE ALARM DEMOLITION PLAN - BASEMENT
SCALE: 1/8" = 1'-0"

FIRE ALARM DEMOLITION PLAN - GROUND FLOOR
SCALE: 1/8" = 1'-0"

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UNIVERSITY OF NORTHERN COLORADO (UNC)
STATE PROJECT # 2017-049P22
GRAY HALL MECHANICAL SYSTEMS REPLACEMENT & LIFE SAFETY UPGRADES
1813 8th Ave., Greeley, Co 80631

SBEC Project #:	230026
Scale:	AS SHOWN
Drawn By:	
Designed By:	AR
Checked By:	JE

Issued For:	Date:
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POWER AND SYSTEMS DEMOLITION PLAN - BASEMENT AND GROUND FLOOR

ED401

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GENERAL NOTES

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DEMOLITION KEY NOTES

(1) XXX



POWER AND SYSTEMS DEMOLITION PLAN - FIRST FLOOR

SCALE: 1/8" = 1'-0"

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POWER AND SYSTEMS DEMOLITION PLAN - FIRST FLOOR

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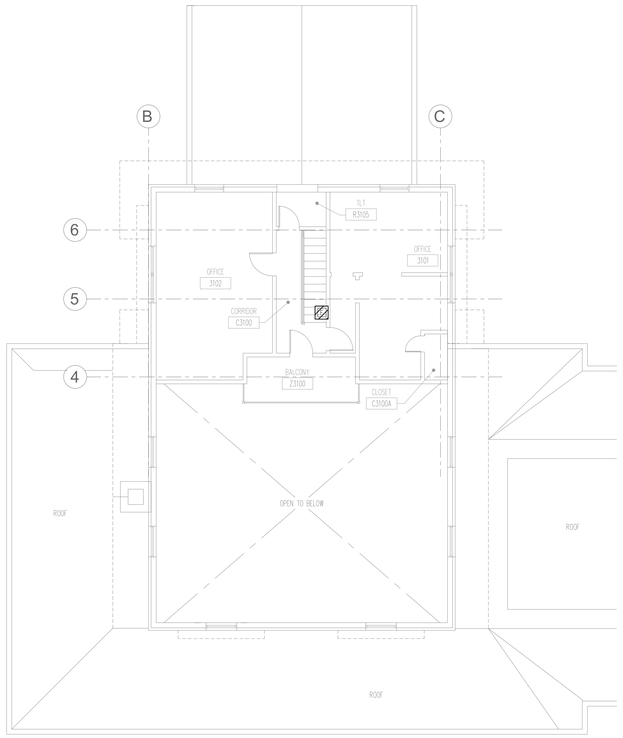
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GENERAL NOTES

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DEMOLITION KEY NOTES

(1) XXX



POWER AND SYSTEMS DEMOLITION PLAN - SECOND FLOOR

SCALE: 1/8" = 1'-0"

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POWER AND SYSTEMS DEMOLITION PLAN - SECOND FLOOR

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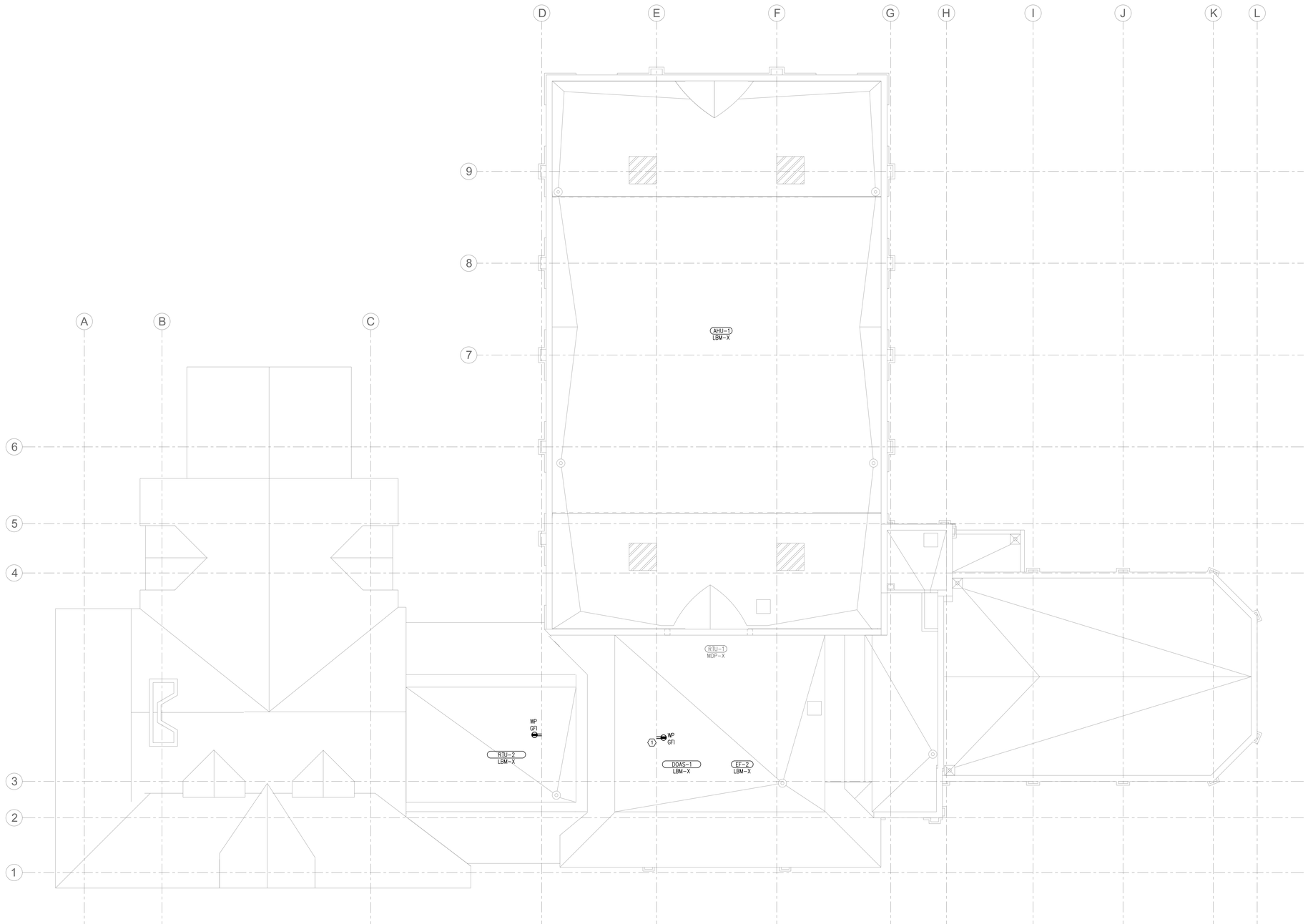
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GENERAL NOTES

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KEY NOTES

(1) LOCATE ROOFTOP RECEPTACLE ABOVE WALL BELOW. PENETRATE ROOF DIRECTLY FROM CONDUIT RUN FROM WALL.



POWER AND SYSTEMS PLAN - ROOF

SCALE: 1/8" = 1'-0"

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POWER AND SYSTEMS DEMOLITION PLAN - ROOF

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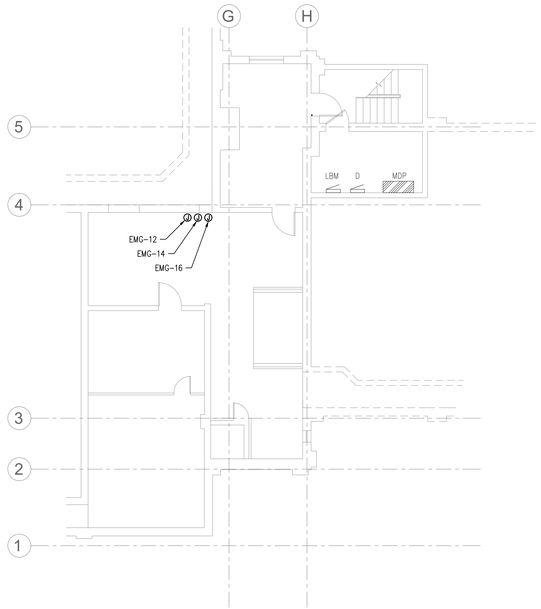
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GENERAL NOTES

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NEW WORK KEY NOTES

(1) PROVIDE LOCAL DISCONNECT SWITCH AND CONNECT TO NEW DRY-PIPE AIR COMPRESSOR. FOR PANEL LOCATION, REFER TO FIRST FLOOR ELECTRICAL PLAN.



PARTIAL ELECTRICAL PLAN - BASEMENT

SCALE: 1/8" = 1'-0"

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GRAY HALL MECHANICAL SYSTEMS REPLACEMENT & LIFE SAFETY UPGRADES

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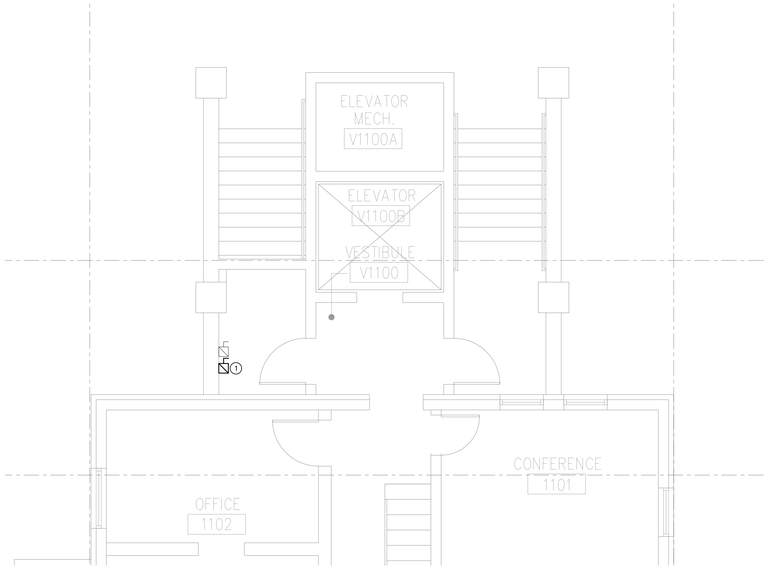
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PARTIAL ELECTRICAL PLAN - BASEMENT

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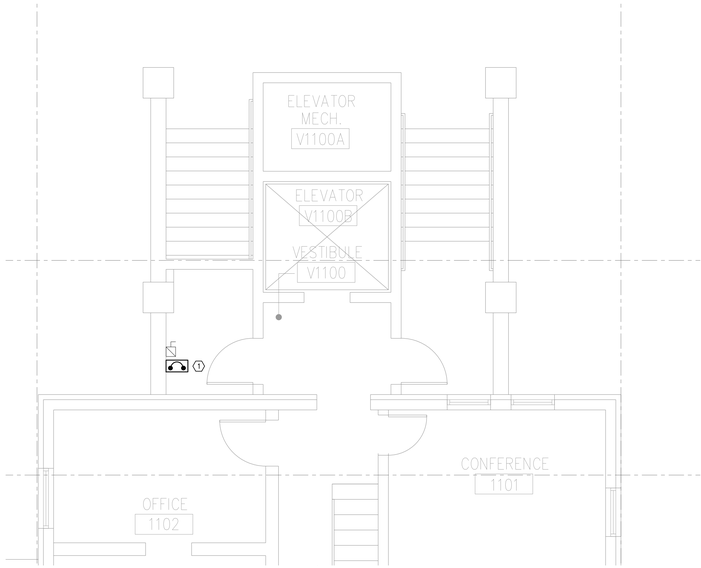
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PARTIAL ELECTRICAL DEMOLITION PLAN - GROUND FLOOR

SCALE: 1/4" = 1'-0"

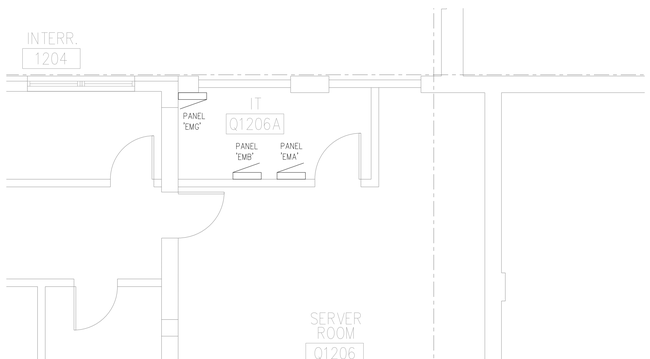
NORTH



PARTIAL ELECTRICAL PLAN - GROUND FLOOR

SCALE: 1/4" = 1'-0"

NORTH



PARTIAL ELECTRICAL PLAN - GROUND FLOOR

SCALE: 1/4" = 1'-0"

NORTH

GENERAL NOTES

1. FOR GENERAL NOTES, REFER TO SHEET E000.

DEMOLITION KEY NOTES

1. REMOVE EXISTING DISCONNECT SWITCH SERVING ELEVATOR AND MAKE SAFE FOR CONNECTION TO NEW ELEVATOR DISCONNECT BREAKER.

NEW WORK KEY NOTES

1. PROVIDE A NEW BUSSMAN ELEVATOR DISCONNECT BREAKER WITH SHUNT TRIP. RECONNECT FEEDERS AS PRACTICAL. PROVIDE CONDUIT MODIFICATIONS AS REQUIRED. WHERE CONDUCTORS ARE INSUFFICIENT TO REACH TERMINATIONS, PROVIDE REPLACEMENT CONDUCTORS.

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ELECTRICAL PLANS - GROUND FLOOR

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POWER AND SYSTEMS PLAN - GROUND FLOOR

SCALE: 1/8" = 1'-0"

NORTH

FIRE ALARM PLAN - BASEMENT

SCALE: 1/8" = 1'-0"

GENERAL NOTES

1. FOR GENERAL NOTES, REFER TO SHEET E000.

KEY NOTES

(1) PROVIDE A REMOTE FIRE ALARM SYSTEM ANNUNCIATOR PANEL WITHOUT SYSTEM RESET CAPABILITY AT LOCATION INDICATED.

(2) PROVIDE GRAPHIC MAP NEAR LOCATION SHOWN. COORDINATE LOCATION WITH AHJ.

(3) PROVIDE SUPERVISION OF WATER FLOW AND TAMPER SWITCHES WITH MONITOR MODULES AS COORDINATED WITH FIRE PROTECTION VENDOR.

(4) PROVIDE WEATHER-RESISTANT HORN/STROBE AT EXTERIOR FIRE DEPARTMENT CONNECTION LOCATION. MOUNT DEVICE AT 10 FT ABOVE GRADE AND IN VIEW OF FIRE DEPARTMENT ACCESS. COORDINATE LOCATION WITH FIRE PROTECTION VENDOR.

(5) PROVIDE ELEVATOR RECALL TO PRIMARY/ALTERNATE FLOOR UPON ALARM OF THIS DETECTOR.

(6) PROVIDE AN AIR SAMPLING SMOKE DETECTOR LOCATED WITHIN THE ROOM NOTED WITH AIR SAMPLE PIPING TO PROVIDE SMOKE DETECTION AT THE BOTTOM AND TOP OF ELEVATOR SHAFT AND RECALL ELEVATOR(S) UPON ALARM OF THIS DETECTOR TO PRIMARY/ALTERNATE FLOOR. THE AIR SAMPLE PIPING SHALL RETURN TO THE DETECTOR UNIT SUCH THAT THE END CAP PORT MAY BE UTILIZED FOR TESTING OF SMOKE TRANSPORT LOCAL TO THE DETECTOR. COORDINATE WITH FIRE PROTECTION VENDOR FOR SPRINKLERS WITHIN THE SHAFT.

(7) PROVIDE MONITOR MODULES AS REQUIRED AND/OR INTERFACE CONNECTION TO BUILDING FIRE ALARM SYSTEM TO MONITOR ASSOCIATED TROUBLE, SUPERVISORY AND ALARM STATUS OF ASSD UNIT.

(8) PROVIDE A LINEAR HEAT DETECTOR UNIT IN THE ROOM NOTED. PROVIDE LINEAR HEAT CABLE WITHIN 18 INCHES OF EACH SPRINKLER HEAD AT THE BOTTOM AND TOP OF THE ELEVATOR SHAFT. THE LINEAR HEAT DETECTION CABLING SHALL RETURN TO THE DETECTOR UNIT SUCH THAT THE END MAY BE UTILIZED FOR TESTING THE HEAT DETECTION CABLING LOCAL TO THE DETECTOR. PROGRAM FIRE ALARM SYSTEM TO ACTIVATE THE SHUNT TRIP OF CIRCUIT BREAKERS ASSOCIATED WITH THIS ELEVATOR VIA CONTROL MODULE AT THE BREAKERS. COORDINATE RESPONSE TIME INDEX RATING OF HEAT DETECTOR CABLE TO SPRINKLER HEAD RATING WITH FIRE PROTECTION VENDOR.

(9) PROVIDE MONITOR MODULES AS REQUIRED AND/OR INTERFACE CONNECTION TO BUILDING FIRE ALARM SYSTEM TO MONITOR ASSOCIATED TROUBLE, SUPERVISORY AND ALARM STATUS OF LINEAR HEAT DETECTOR UNIT.

(10) PROVIDE A FIXED TEMPERATURE, QUICK RESPONSE HEAT DETECTOR WITHIN 18 INCHES OF EACH SPRINKLER HEAD WITHIN THE ELEVATOR MACHINE ROOM/SPACE AND CONNECT TO THE SHUNT TRIP OF CIRCUIT BREAKERS ASSOCIATED WITH THIS ELEVATOR VIA CONTROL MODULE. COORDINATE RESPONSE TIME INDEX RATING OF HEAT DETECTOR TO SPRINKLER HEAD RATING WITH FIRE PROTECTION VENDOR.

(11) PROVIDE CONTROL MODULE AT SHUNT TRIP LOCATION TO BE ACTIVATED UPON ALARM OF HEAT DETECTOR UNIT(S) FOR DETECTION AT THE BOTTOM AND/OR TOP OF THE ELEVATOR SHAFT OR IN THE ASSOCIATED ELEVATOR MACHINE ROOM.

(12) PROVIDE CONTROL MODULES FOR ELEVATOR PRIMARY FLOOR RECALL, ALTERNATE FLOOR RECALL, AND FIRE FIGHTER'S HAT INDICATOR. PRIMARY FLOOR SHALL BE LEVEL 01, ALTERNATE FLOOR SHALL BE LEVEL 02. CONNECT TO ELEVATOR CONTROLLER PER ELEVATOR INSTALLER'S INSTRUCTIONS.

(13) PROVIDE MONITOR MODULE AS REQUIRED FOR SUPERVISION OF THE SHUNT TRIP POWER CIRCUIT AT SOURCE.

(14) PROVIDE DISCONNECT OF THE INDICATED SPEAKER ONLY UPON ACTIVATION OF THE VOICE EVACUATION MICROPHONE/HANDSET.

(15) PROVIDE SUPERVISION OF LOW AIR PRESSURE SWITCH ASSOCIATED WITH DRY PIPE SYSTEM WITH MONITOR MODULE. COORDINATE WITH FIRE PROTECTION VENDOR.

(16) PROVIDE SHUT DOWN OF AIR HANDLING UNIT UPON ALARM OF THIS DETECTOR VIA CONTACTS WITHIN THE DUCT DETECTOR. CONNECT TO AIR HANDLING UNIT CONTROLLER PER MANUFACTURER'S INSTRUCTIONS.

(17) PROVIDE A REMOTE KEYED TEST SWITCH AND INDICATOR LIGHT ON THE WALL NEAR DUCT DETECTOR AND MOUNT AT 7'-0" AFF. LABEL TEST SWITCH WITH DETECTOR ADDRESS, AHU NAME AND EITHER 'SUPPLY' OR 'RETURN'.

(18) PROVIDE MONITOR MODULES CONNECTED TO FIRE SUPPRESSION CLEAN AGENT PANEL TO MONITOR FOR RELEASE OF AGENT, SUPERVISORY AND TROUBLE.

(19) PROVIDE DETECTION AND SUPERVISION OF PRE-ACTION WATER FLOW AND TAMPER SWITCHES PER FIRE PROTECTION DESIGN. SYSTEM SHALL BE A DOUBLE-INTERLOCKED DESIGN. COORDINATE PROGRAMMING WITH FIRE PROTECTION.

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POWER AND SYSTEMS PLAN - BASEMENT AND GROUND FLOOR

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GENERAL NOTES

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KEY NOTES

(1) PROVIDE ELEVATOR RECALL TO PRIMARY/ALTERNATE FLOOR UPON ALARM OF THIS DETECTOR.

(2) PROVIDE FIRE ALARM/VOICE EVACUATION CONTROL PANEL (FACP) IN LOCATION SHOWN. COORDINATE WITH AUTHORITY HAVING JURISDICTION FOR APPROVAL OF FINAL LOCATION PRIOR TO ROUGH-IN.

(3) PROVIDE GRAPHIC MAP NEAR LOCATION SHOWN. COORDINATE LOCATION WITH AHJ.



POWER AND SYSTEMS PLAN - FIRST FLOOR

SCALE: 1/8" = 1'-0"

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UNIVERSITY OF NORTHERN COLORADO (UNC)
STATE PROJECT # 2017-049P22
GRAY HALL MECHANICAL SYSTEMS REPLACEMENT & LIFE SAFETY UPGRADES
1813 8th Ave., Greeley, Co 80631

SBEC Project #:	230026
Scale:	AS SHOWN
Drawn By:	
Designed By:	AR
Checked By:	JE

Issued For:	Date:
100% DD	03/11/2025

POWER AND SYSTEMS PLAN - FIRST FLOOR

E403

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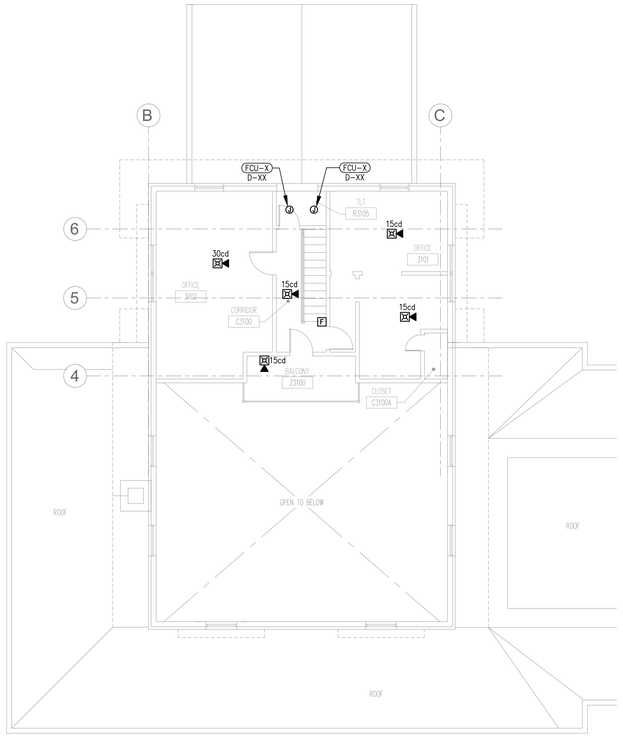
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GENERAL NOTES

1. FOR GENERAL NOTES, REFER TO SHEET E000.

KEY NOTES

(1) XXX



POWER AND SYSTEMS PLAN - SECOND FLOOR

SCALE: 1/8" = 1'-0"

NORTH

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POWER AND SYSTEMS PLAN - SECOND FLOOR

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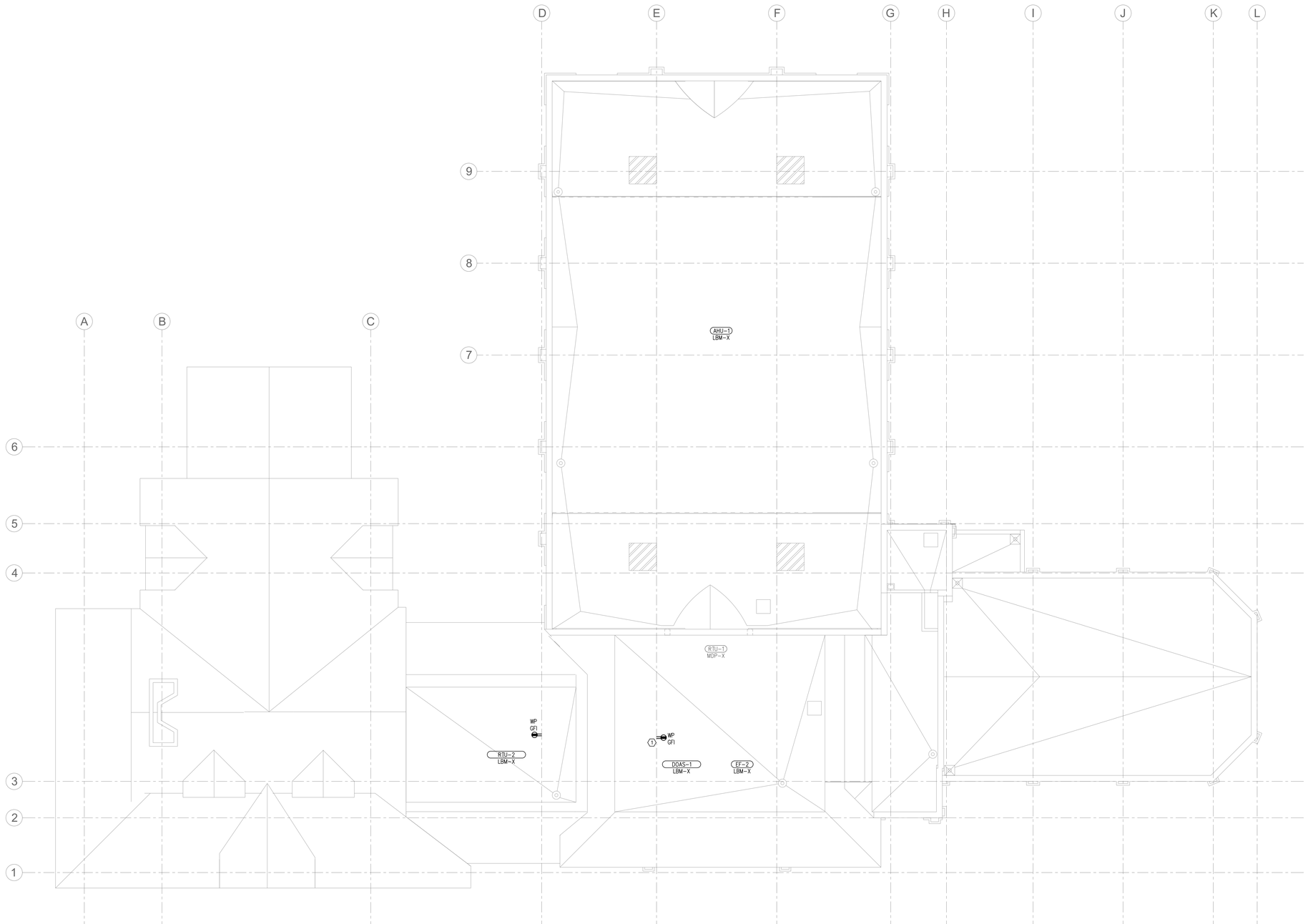
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KEY NOTES

(1) LOCATE ROOFTOP RECEPTACLE ABOVE WALL BELOW. PENETRATE ROOF DIRECTLY FROM CONDUIT RUN FROM WALL.



POWER AND SYSTEMS PLAN - ROOF

SCALE: 1/8" = 1'-0"

NORTH

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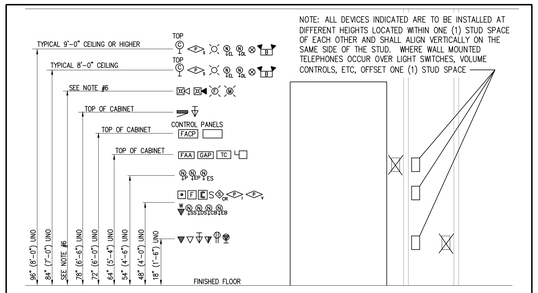
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POWER AND SYSTEMS PLAN - ROOF

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NOTES:

1. HEIGHTS SHOWN ARE TYPICAL TO CENTERLINE OF BOX UNLESS NOTED OTHERWISE, I.E. "TOP OF DEVICE" OR "TOP."
2. WHEREVER DEVICES ARE NOTED TO BE ABOVE DOORS, DEVICE SHALL BE CENTERED BETWEEN TOP OF DOOR TRIM AND CEILING LINE.
3. MOUNTING HEIGHTS SHOWN ON ARCHITECTURAL ELEVATIONS SHALL GOVERN OVER THOSE SHOWN ABOVE.
4. OUTLETS ABOVE COUNTERS SHALL BE MOUNTED 8" ABOVE COUNTER OR 4" ABOVE BACKSPLASH.
5. MOUNTING HEIGHTS SHALL ALSO COMPLY WITH ADA REQUIREMENTS.
6. MOUNTING HEIGHT OF WALL MOUNTED VISIBLE STROBES AND COMBINATION AUDIBLE/VISIBLE ALARM DEVICES SHALL BE CONSISTENT AND IN ACCORDANCE WITH NFPA REQUIREMENT THAT BOTTOM OF STROBE LENS BE NOT LESS THAN 80" AFF AND NOT MORE THAN 96" AFF, WHILE NOT LESS THAN 6" BELOW FINISHED CEILING. COORDINATE DEVICE BACK BOX MOUNTING HEIGHT WITH DEVICE MANUFACTURER TO COMPLY WITH NFPA.

DEVICE MOUNTING HEIGHT DETAIL

SCALE: NONE

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ELECTRICAL DETAILS

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